



Imagery Data Assistant

AI-Powered Geospatial Analysis & Visualization

Version 0.1.0

Platform ArcGIS Maps SDK for JavaScript 5.0 + React

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Table of Contents

1. [Overview](#)
2. [Architecture](#)
3. [Technology Stack](#)
4. [Agent Capabilities](#)
5. [Example Workflows](#)
6. [Getting Started](#)

1. Overview

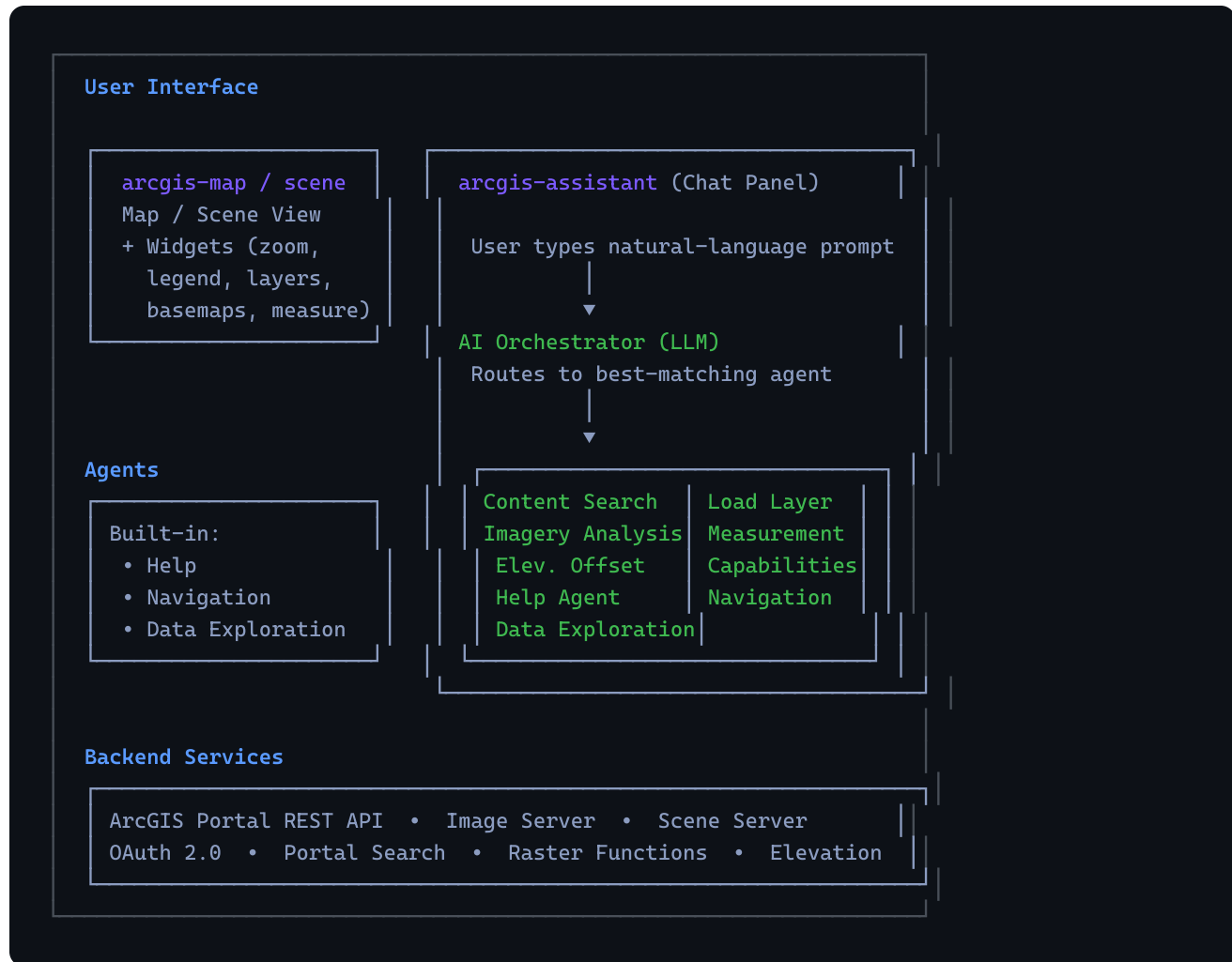
The **Imagery Data Assistant** is a conversational AI-powered web application for geospatial data discovery, visualization, and analysis. Users interact with a natural-language chat interface to search for imagery and geospatial layers, load them onto a 2D or 3D map, apply raster analysis functions, take measurements, and adjust 3D elevation offsets — all without writing code or navigating complex menus.

Key Idea: Instead of clicking through toolbars and dialogs, users simply describe what they want in plain English. The assistant's AI orchestrator routes each request to the appropriate specialized agent, which executes the task and reports results back in the chat.

Core Capabilities at a Glance

- **Content Search & Discovery** — Search My Content, My Org, ArcGIS Online, or Living Atlas for layers by keyword
- **Layer Loading** — Load any ArcGIS-supported layer type from a URL, item ID, or keyword search
- **Imagery Analysis** — Apply stretches, color ramps, NDVI, hillshade, slope, and server-side raster functions
- **Measurement Tools** — Distance, area, and elevation profile with configurable units
- **3D Elevation Offset** — Auto-fix or manually adjust floating/underground 3D layers
- **Navigation & Help** — Zoom to locations, change basemaps, get contextual help

2. Architecture



How It Works

1. The user types a natural-language prompt in the **arcgis-assistant** chat panel.
2. The **AI Orchestrator** (powered by an LLM via LangChain/LangGraph) analyzes the prompt and routes it to the most relevant agent based on each agent's description.
3. The selected **agent** executes a LangGraph state graph: it extracts intent via fast regex patterns (falling back to LLM when needed), calls ArcGIS APIs, and returns a chat response.
4. Results are displayed in the chat — layers appear on the map, measurement tools activate, analysis renders, etc.

Smart Routing: Each agent includes bail-out logic that detects requests meant for other agents and returns silently. This prevents conflicts when multiple agents receive the same message from the

orchestrator.

3. Technology Stack

Component	Technology
Mapping Engine	ArcGIS Maps SDK for JavaScript 5.0 (@arcgis/core 5.0.3)
Map Web Components	@arcgis/map-components 5.0.3 — arcgis-map, arcgis-scene, arcgis-zoom, arcgis-home, arcgis-legend, arcgis-layer-list, arcgis-basemap-gallery, measurement components
AI Assistant Components	@arcgis/ai-components 5.0.3 — arcgis-assistant, arcgis-assistant-agent, built-in help/navigation/data-exploration agents
UI Design System	Calcite Design System (@esri/calcite-components 5.0.2) — shell, panels, buttons, icons, segmented controls, loaders
AI Orchestration	LangChain LangGraph (@langchain/langgraph 1.1.5) — StateGraph-based agent execution with node/edge workflows
Frontend Framework	React 18.3 with TypeScript 5.9
Build Tool	Vite 7.3 with React plugin
Validation	Zod 4.3 for runtime type checking
Authentication	ArcGIS OAuth 2.0 via IdentityManager

Key ArcGIS Web Components Used

Component	Purpose
<arcgis-map>	2D MapView container
<arcgis-scene>	3D SceneView container
<arcgis-assistant>	AI chat panel with orchestrator
<arcgis-assistant-agent>	Custom agent registration element
<arcgis-assistant-help-agent>	Built-in help / FAQ agent

<arcgis-assistant-navigation-agent>	Built-in map navigation agent
<arcgis-assistant-data-exploration-agent>	Built-in data exploration agent
<arcgis-distance-measurement-2d>	2D distance measurement tool
<arcgis-area-measurement-2d>	2D area measurement tool
<arcgis-direct-line-measurement-3d>	3D line measurement tool
<arcgis-area-measurement-3d>	3D area measurement tool
<arcgis-elevation-profile>	Elevation cross-section tool
<calcite-shell>	Application layout shell

4. Agent Capabilities

The assistant includes **9 agents** — 3 built-in ArcGIS agents and 6 custom agents. The AI orchestrator selects the best agent for each user request based on the agent descriptions.

Help Agent BUILT-IN

Answers general questions about the assistant, its capabilities, and how to use the application. Provides contextual guidance based on the current map state.

Navigation Agent BUILT-IN

Handles map navigation commands: zoom to locations, change basemaps, adjust view extent, go to coordinates, and manage camera position in 3D.

Data Exploration Agent BUILT-IN

Explores data attributes and statistics of layers loaded on the map. Supports filtering, querying, and summarizing feature attributes.

Content Search Agent CUSTOM

Searches for geospatial layers across multiple ArcGIS sources and supports batch loading of results.

Search Scopes

- **My Content** — user's own portal items
- **My Organization** — items shared within the org
- **ArcGIS Online** — public community content
- **Living Atlas** — Esri's curated authoritative datasets
- **All** — searches all scopes simultaneously

Batch Loading

- `add result 3` — add a single result
- `add results 1-5` — add a range
- `add results 1, 3, 7` — add specific items
- `add all results` — add everything from the search

Example Prompts

- `search my content for Phoenix imagery`
- `find elevation data in Living Atlas`
- `from my org, find all layers related to wildfire`

Load Layer Agent CUSTOM

Loads any ArcGIS-supported layer type onto the map from a URL, portal item ID, or keyword search.

Supported Layer Types

Imagery, Feature, Tile, Map Image, Scene, Integrated Mesh, 3D Tiles, Gaussian Splat, Point Cloud, Building Scene, Oriented Imagery, WMS, WFS, WMTS, KML, GeoJSON, CSV, Vector Tile

Smart Features

- Auto-detects layer type from URL pattern or portal item metadata
- Automatically switches to 3D view when loading 3D-only layers
- Supports elevation offset on load (e.g., `load mesh with 50m offset`)
- Auto-zooms to the loaded layer's extent

Example Prompts

- `load world imagery`
- `load https://services.arcgis.com/.../ImageServer`
- `show me the layer with ID a1b2c3d4 ...`

Imagery Analysis Agent CUSTOM

Analyzes and visualizes imagery layers already loaded on the map using renderers, raster functions, and interactive tools.

Capabilities

Category	Options
Stretch Renderers	Standard Deviation, Min-Max, Percent Clip
Color Ramps	Inferno, Viridis, Grayscale
Raster Functions	NDVI, Hillshade, Slope, Aspect, Color Infrared (False Color)
Server-side Functions	Automatically discovers and applies server processing templates
Interactive Tools	Click-to-identify pixel popup, center pixel identification
Export	Screenshot capture (PNG)

Example Prompts

- `apply standard deviation stretch`
- `render with inferno color ramp`
- `apply NDVI`
- `show pixel values at center`
- `list available raster functions`
- `take a screenshot`

Measurement Agent

CUSTOM

Activates interactive measurement tools on the map. Automatically selects the correct 2D or 3D widget based on the current view.

Measurement Types

Type	2D Component	3D Component
Distance	arcgis-distance-measurement-2d	arcgis-direct-line-measurement-3d
Area	arcgis-area-measurement-2d	arcgis-area-measurement-3d
Elevation Profile	arcgis-elevation-profile (works in both)	

Supported Units

- **Linear:** meters, kilometers, feet, miles, yards, nautical miles
- **Area:** square meters, square kilometers, square feet, square miles, acres, hectares

Example Prompts

- `measure distance in kilometers`
- `measure area in acres`
- `show elevation profile`
- `clear measurement`

Elevation Offset Agent CUSTOM

Fixes 3D layers that appear floating above or sunken below the terrain by adjusting their elevation offset. Supports automatic terrain-based correction and manual adjustments.

Modes

- **Auto-fix** — Samples terrain elevation at a 5×5 grid across the layer extent, calculates the optimal offset, and applies it automatically
- **Manual Set** — Sets an absolute offset value (e.g., `set offset to 50 meters`)
- **Relative Adjust** — Adds or subtracts from current offset (e.g., `raise by 10m`)
- **Check** — Reports current offset and Z-range without making changes

Supported 3D Layer Types

Integrated Mesh, Scene Layer, Point Cloud, Building Scene Layer, Gaussian Splat, 3D Tiles

Example Prompts

- `fix the floating mesh`
- `align this layer to the ground`
- `add 16 meters to the layer`
- `lower the mesh by 20 feet`

All Capabilities Agent CUSTOM

Meta-agent that lists all available assistant capabilities. Dynamically discovers all registered agents (both built-in and custom) and formats a human-readable summary. Triggered by prompts like `what can you do?` or `list capabilities`.

5. Example Workflows

Workflow 1: Discover and Analyze Imagery

Scenario: A GIS analyst wants to find and analyze thermal imagery for a city.

1. `search Living Atlas for thermal imagery in Phoenix`
Content Search Agent returns a numbered list of matching imagery layers.
2. `add result 1`
The layer is loaded onto the map and the view zooms to its extent.
3. `apply standard deviation stretch`
Imagery Analysis Agent enhances the contrast for better visualization.
4. `render with inferno color ramp`
A heat-style color ramp is applied to highlight temperature differences.
5. `enable click-to-identify popup`
Clicking anywhere on the imagery now shows pixel/band values.
6. `take a screenshot`
The current view is exported as a PNG image in a new browser tab.

Workflow 2: Batch Load Organizational Content

Scenario: A project manager wants to load all team layers for a project area.

1. `from my content, find all layers related to Phoenix`
Content Search Agent searches the user's portal content and returns all matching items.
2. `add all results`
All results are loaded concurrently onto the map. If any require 3D, the view auto-switches to a Scene before loading.
3. `measure area in acres`
The area measurement tool activates for the user to draw a polygon.

Workflow 3: Load and Fix a 3D Mesh

Scenario: A drone survey produced a 3D mesh that floats above the terrain.

1. `load https://tiles.arcgis.com/ ... /3DTilesServer/tileset.json`
Load Layer Agent detects a 3D Tiles layer, auto-switches to 3D SceneView, and loads it.
2. `fix the floating mesh`
Elevation Offset Agent samples terrain at 25 points across the layer extent, calculates the ground-level offset, and applies the correction.
3. `raise by 2 meters`
Fine-tunes the offset with a relative +2m adjustment.
4. `measure distance in feet`
3D direct-line measurement tool activates in feet for verification.
5. `elevation profile`
The elevation profile tool lets the user draw a line across the mesh to verify alignment.

Workflow 4: Multi-Source Comparison

Scenario: Comparing elevation data from different sources.

1. `search ArcGIS Online for NAIP imagery Arizona`
Finds NAIP aerial imagery layers.
2. `add result 1`
Loads the NAIP imagery as a base reference layer.
3. `apply NDVI`
Applies Normalized Difference Vegetation Index to visualize vegetation health.
4. `find elevation data in Living Atlas`
Searches for DEM / terrain layers in Esri's curated collection.
5. `add results 1-3`
Loads the top three elevation datasets for comparison.
6. `apply hillshade to the elevation layer`
Renders terrain shading for better visual interpretation.

Workflow 5: Quick Q&A and Exploration

Scenario: A new user learning the application.

1. `what can you do?`
All Capabilities Agent lists every available function with descriptions.
2. `zoom to San Francisco`
Navigation Agent flies the map to San Francisco.
3. `load world imagery`
Loads the Esri World Imagery basemap layer.
4. `how far is it from here to Oakland?`
Measurement Agent activates the distance tool for the user to measure.

6. Getting Started

Prerequisites

- Node.js 18+ and npm
- An ArcGIS Online or ArcGIS Enterprise account
- An OAuth 2.0 Application ID registered in ArcGIS (for portal content access)

Setup

Environment Variables Required:

VITE_ARCGIS_OAUTH_APP_ID — Your ArcGIS OAuth application ID (required)

VITE_ARCGIS_PORTAL_URL — Portal URL (optional, defaults to <https://www.arcgis.com>)

VITE_WEBMAP_ITEM_ID — Existing web map ID (optional, auto-creates if not set)

1. Clone the repository and navigate to `Apps/imagery-assistant`
2. Create a `.env` file with your OAuth App ID
3. Install dependencies: `npm install`
4. Start the dev server: `npm run dev`
5. Open the URL shown in your browser and sign in with your ArcGIS account

Build for Production

Run `npm run build` to produce an optimized production build in the `dist/` folder.